

Vivian Mak

Robotics Software Engineer

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Education

Olin College of Engineering

BS in Robotics Engineering | GPA: 3.83

May 2027

Needham, MA

- **Relevant Coursework:** Computational Robotics and Dynamics (TA), Software Design, Principles of Integrated Engineering, Data Structures and Algorithms, Image Processing, Robot Kinematics and Dynamics

Shanghai University of Finance and Economics (Study Abroad)

May 2026

Experience

Amazon Robotics (Senior Capstone)

Sept 2026 - Present

Perception Researcher | Technical Lead

- Leading a team of 5 engineers to deliver a low-cost, **universal-robot sensing platform**, by owning timeline planning, overseeing integration, and communicating with executive sponsor
- Architecting a perception (detection, tracking, prediction) pipeline for **dynamic environments**, across various weather conditions, and offline operation

Eyebot

Jun 2026 - Aug 2026

R&D Computer Vision Intern

- Built a **Kalman-filter** based gaze tracking system with adaptive smoothing to improve tracking stability and driving a state machine to recognize passing glances
- Optimized a **vision transformer** on a self-curated dataset for glasses detection by training models over a parameter sweep, achieving a model with 95% accuracy (~90% error reduction) and a 360ms inference time (x4.6 latency reduction)

Olin Underwater Robotics Lab

Jan 2025 - Present

Student Researcher

- Researching and developing a **3D reconstruction** pipeline to analyze coral temporal dynamics with **plenoptic camera**
- Conducting camera calibration and controlled **underwater experiments**, defining quantitative metrics for 3D reconstruction accuracy informed by an ongoing literature review, and preparing an academic publication on results

Stealth Startup

May 2025 - Aug 2025

Testing and Validation Intern

- Conducted **engineering validation testing** with reliability modeling scripts to verify 4 major performance claims and predict 33 component lifetime failure rates, directly guiding design direction and service strategy

Olin Electric Motorsports | Formula SAE Electric

Sept 2023 - Jun 2025

Lead Battery Engineer | Mentor

- Lead a team of 10 engineers to design and manufacture an EV battery, by organizing team meetings, coordinating integration with electrical systems, managing project timelines, mentoring new members, ensuring safety (**HV Trained**)
- Specified, designed, and manufactured a **428V battery pack** using **SOLIDWORKS**, focusing on optimizing mechanical packaging, reducing weight by 20% with respect to previous iteration

Projects

Pose Graph Optimization for SLAM

Oct 2025 - Dec 2025

- Built a LiDAR-odometry **SLAM** pipeline in **C++** with scan matching, motion estimation, and pose-graph optimization

Fleet Robotics

Nov 2024 - Dec 2024

- Developed a **visual odometry** pipeline for a **decentralized multi-agent** ROS2 fleet planner to improve pose estimation

Robot State Estimation and Localization

Oct 2024

- Implemented a **particle filter** in **ROS2 Python** utilizing **LiDAR** data to perform global state estimation and localization

Skills

Software: Python, C, C++, Arduino, MATLAB, Git, Linux, ROS2, Gazebo, Rviz2

Mechanical: SOLIDWORKS, OnShape, CNC Router, CNC Mill, 3D Printing

Languages: English, Mandarin, Cantonese

Conferences: Northeast Robotics Colloquium (2025), IEEE OCEANS (2026, poster presentation)